UNIT IV

17. Prove that $x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} (-1)^n \frac{\cos nx}{n^2}$, $-\pi < x < \pi$.

Hence show that :

(a) $\sum \frac{1}{n^2} = \frac{\pi^2}{6}$

(b) $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \ldots + \frac{\pi^2}{12}$.

Or

18. Express $f(x) = x$ as a half-range cosine series in $0 < x < 2$.

UNIT V

19. Prove that in $0 < x < l$,

$$x = \frac{1}{2} - \frac{4l}{\pi^2}\left(\cos\frac{\pi x}{l} + \frac{1}{3^2}\cos\frac{3\pi x}{l} + \frac{1}{5^2}\cos\frac{5\pi x}{l} + \ldots\right)$$

and deduce that $\frac{1}{1^4} + \frac{1}{3^4} + \frac{1}{5^4} + \ldots + \frac{\pi^4}{96}$.

Or

20. Obtain the first three coefficients for the Fourier cosine series for y, where y is given in the following table :

x:	0	1	2	3	4	5
y:	4	8	15	7	6	2

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2π0/n

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2019.

Third Semester

MATHEMATICS – III

(Common to All Branches)

(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. If $f(z)$ is an analytic function with constant modulus then show that $f(z)$ is constant.

2. If $f(z) = u + iv$ be an analytic function in some region of the z-plane then prove that u satisfies the Laplace's equation in two variables.

3. Find the bilinear transformation which maps the points $z = 1, i, -1$ on to the points $w = i, 0, -i$.

4. Show that a bilinear transformation preserves cross-ratio of four points.

5. Evaluate $\int_C (z-a)^{-1}\, dz$, where C is a simple closed curve and the point $z = a$ is inside C.

6. Determine the poles of the function $f(z)=\frac{z^2}{(z-1)^2(z+2)}$.

7. Obtain the Fourier coefficient a_0 for $f(x)=e^{-x}$ in $0<x<2\pi$.

8. Obtain the Fourier coefficient a_n for $f(x)=\pi x$ in $0\le x\le 2$.

9. State Parseval's formula.

10. Explain the complex form of Fourier series.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the Units, choosing ONE full question from each Unit.

UNIT I

11. Prove that the function $f(z)$ defined by $f(z)=\frac{x^2(1+j)-y^3(1-i)}{x^2+y^2}(z\neq 0)$, $f(0)=0$ is continuous and the Cauchy-Riemann equations are satisfied at the origin, ye $f'(0)$ does not exist.

Or

12. If $f(z)$ is a regular function of z, then prove that $\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2=4|f'(z)|^2$.

UNIT II

13. Show that, under the transformation $w=\frac{1}{z}$, circle $x^2+y^2-6x=0$ is transformed into a straight line in the w-plane.

Or

14. Expand $f(z)=\frac{1}{(z-1)(z-2)}$ in the region

(a) $|z|<1$

(b) $1<|z|<2$

(c) $|z|>2$.

UNIT III

15. Using Cauchy's integral formula, evaluate $\int_C \frac{e^{2z}dz}{(z-1)(z-2)}$ where C is the circle $|z|=3$.

Or

16. By integrating around a unit circle, evaluate $\int_0^{2\pi}\frac{\cos 3\theta}{5-4\cos\theta}d\theta$.

UNIT V

19. (a) Find the R.M.S. value of $f(x) = x^2$ in $-\pi \le x \le \pi$. (5)

(b) Find the complex form of the Fourier series of the periodic function $f(x) = \sin x$ in $0 < x < \pi$. (6)

Or

20. Find the Fourier series upto 3rd harmonic for the function $y = f(x)$ define in $(0, \pi)$ from the table

x	0	$\frac{\pi}{6}$	$\frac{2\pi}{6}$	$\frac{3\pi}{6}$	$\frac{4\pi}{6}$	$\frac{5\pi}{6}$	π
$f(x)$	2.34	2.2	1.6	0.83	0.51	0.88	1.19

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B.Tech. DEGREE EXAMINATION,
MAY 2019.

Third Semester

Civil Engineering

MATHEMATICS — III

(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. State the Cauchy-Riemann equation in polar form.

2. Show that the function $x^4 - 6x^2y^2 + y^4$ is harmonic.

3. Define bilinear transformation.

4. State Taylor's theorem.

5. Write the Cauchy's integral formula.

6. Evaluate $\int_c \frac{dz}{2z+1}$ where c is $|z| = 1$.

7. Pick out the even function and justify your answer $x^2, \sin x, 1 + x, e^x$.

8. Write the formulae for Fourier constants for $f(x)$ in the interval $(-\pi, \pi)$.

9. What is the RMS value of $f(x)$ in $(-\pi, \pi)$.

10. State Parseval's theorem on Fourier coefficients.

SECTION B — ($5 \times 11 = 55$ marks)

Answer ALL the units, choosing ONE full questions from each Unit.

UNIT I

11. Verify Cauchy-Riemann equations for the function $f(z) = z^3$.

Or

12. Prove that $u = 2x - x^3 + 3xy^2$ is harmonic. Find the corresponding analytic function.

UNIT II

13. Find the bilinear transformation which maps the points $z_1 = 2$, $z_2 = i$ and $z_3 = -2$ into the points $w_1 = 1, w_2 = i$ and $w_3 = -1$.

Or

14. (a) Expand $z^{-1}e^{-2z}$ in Laurent's series about $z = 0$. (5)

(b) Define the types of singularities and give an example to each. (6)

UNIT III

15. Using Cauchy's integral formula, find the value of $\int_c \frac{z+4}{z^2+2z+5} dz$ where c is the circle $|z+1-i| = 2$.

Or

16. Prove that $\int_0^{2\pi} \frac{d\theta}{2+\cos\theta} = \frac{2\pi}{\sqrt{3}}$.

UNIT IV

17. Find the Fourier series for the function $f(x) = e^x$ defined in $(-\pi, \pi)$.

Or

18. Obtain the half range cosine series for $f(x) = x$ in $(0, \pi)$.

UNIT IV

17. Find the Fourier series for the function $f(x) = e^x$ defined in $(-\pi, \pi)$.

Or

18. Obtain the half range cosine series for $f(x) = x$ in $(0, \pi)$ and deduce that the sum of the series $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots..$

UNIT V

19. Find the complex form of Fourier series of $f(x) = e^{ax}$ $(-\pi < x < \pi)$ in the form $e^{ax} = \frac{\sin h a\pi}{\pi} \sum_{-\infty}^{+\infty} (-1)^n \frac{a+in}{a^2+n^2} e^{inx}$.

Or

20. Compute the first two harmonics of the Fourier series of $f(x)$ from the following tables.

x	30	60	90	120	150	180	210
$f(x)$	2.34	3.01	3.68	4.15	3.69	2.2	0.83
x	240	270	300	330	360		
$f(x)$	0.51	0.88	1.09	1.19	1.64		

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B.Tech. DEGREE EXAMINATION,
NOVEMBER/DECEMBER 2018.

Third Semester

Civil Engineering

MATHEMATICS – III

(From 2013–2014 Onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Show that the function $f(z) = e^x \cos y$ in harmonic.

2. Show that an analytic function with constant real part is a constant.

3. Find the image of the circle $|z| = 2$ under the transformation $w = 3z$.

4. Find the invariant points of the transformation $w = \frac{(2z+4i)}{iz+1}$.

5. Using Cauchy's integral formula evaluate $\int_C \frac{z+1}{z^3-2z^2}dz$ where C is the circle $|z-2-i|=2$.

6. Using Cauchy's residue theorem evaluate $\int_C \frac{dz}{(z^2+9)^3}$ where C is $|z-i|=3$.

7. If $f(x)=x\sin x$, $0\le x\le 2\pi$ find the Fourier coefficient a_0.

8. Obtain the Fourier coefficient an for the function $f(x)=x^2\ -\pi<x<\pi$.

9. State Parsevals identity.

10. Write down the Fourier coefficients a_0, a_n and b_n of the function $y=f(x)$ in $(0, 2\pi)$ with respect to harmonic analysis.

SECTION B — $(5 \times 11 = 55$ marks)

Answer All the units, choosing ONE from each unit.

UNIT I

11. Find an analysis function $w=u+iv$ if $u=e^x(x\sin y+y\cos y)$. Hence find v.

Or

12. Find the equation of the orthogonal trajectories of the family of curves given by $3x^2y+2x^2-y^3-2y^2=a$.

UNIT II

13. Find the bilinear transformation that maps the points $1+i$, $-i$, $2-i$ of the z-plane into the points $0, 1, i$ of the w-plane.

Or

14. Find the Image of the rectangular region in the z-plane bounded by the lines $x=0$, $y=0$, $x=2$ and $y=1$ under the transformation $w=ze^{i\pi/4}$.

UNIT III

15. If $f(a)=\int_c \frac{3z^2+7z+1}{z-a}dz$ where C is the circle $|z|=2$ find the values of $f(3)$ $f'(1-i)$ and $f''(1-i)$.

Or

16. Using contour integration, evaluate $\int_0^{2\pi}\frac{d\theta}{(a+b\cos\theta)}\ (a>b>0)$.

UNIT IV

→ mid, mod

17. Expand $f(x) = (\pi - x)^2$ in $(-\pi, \pi)$ as a Fourier series.

mid model

Or

18. Find the half range sine series of $f(x) = x\cos x$ in $(0, \pi)$.

UNIT V

, model

19. Find the Fourier series expansion of period 2π for the function $y = f(x)$ which is defined in $(0, 2\pi)$ by means of the table of values given below. Find the series upto the third harmonic.

X	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
Y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Or

→ model

20. Find the complex form of Fourier series of $f(x) = e^{ax} (-\pi < x < \pi)$ in the form $e^{ax} = \frac{\sinh a\pi}{\pi} \sum_{-\infty}^{\infty} (-1)^n \frac{a + in}{a^2 + n^2} e^{inx}$ and hence prove that $\frac{\pi}{a \sinh a\pi} = \sum_{-\infty}^{\infty} \frac{(-1)^n}{n^2 + a^2}$.

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B.Tech. DEGREE EXAMINATION, NOVEMBER/DECEMBER 2017.

Third Semester

Civil Engineering

MATHEMATICS – III

(From 2013-14 onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL the questions.

1. Define harmonic function. → Model, mid,

2. Test whether the function $f(z) = x^2 + iy^2$ is an analytic or not. → Unit 1, Model, mid

3. Define bilinear transformation. → mid,

4. What are types of singularities? → mid,

5. State Jordan's lemma. → No

mid, model

6. Evaluate, using Cauchy s integral formula $\int_C \frac{zdz}{z-2}$.

Where C is the Circle $|z|=1$.

mid, model

7. Define even function with an example.

remodel

8. State Dirichiet's conditions for Fourier series.

model

9. Define Root mean square value of a function $f(x)$.

model

10. State Parseval's theorem on Fourier coefficients.

SECTION B — (5 × 11 = 55 marks)

Answer ALL the units, choosing ONE full question from each Unit.

UNIT I

unit, mid, model

11. (a) Test the analyticity of the function $f(z)=e^x(\cos y+i\sin y)=e^z$. (5)

(b) Show that $f(z)=\sin z$ is an analytic function.

mid, unit, model (6)

Or

12. Show that the function $u=2xy+3y$ is harmonic and find the corresponding analytic function. Find its conjugate.

model, unit.

UNIT II

13. Find the bilinear transformation which maps the points $z_1=2$, $z_2=i$ and $z_3=-2$ into the points $w_1=1$, $w_2=i$ and $w_3=-1$. unit 1, model, mid.

Or

14. Obtain Taylor's series to represent the function $\frac{z^2-1}{(z+2)(z+3)}$ in the region $|z|<2$. mid, model

UNIT III

15. Evaluate $\int_0^{2\pi} \frac{d\theta}{5-4\sin\theta}$. mid, model

Or

16. (a) Evaluate, using Cauchy's integral formula $\frac{1}{2\pi i}\int_C \frac{z^2+5}{z-3}dz$ on the circle (i) $|z|=4$ and

model

(ii) $|z|=1$. (6)

remodel, mid.

(b) Find the residue of $\frac{z+2}{(z-2)(z+1)^2}$. (5)

18. Find the half range cosine series for the function $f(x)=(x-1)^2$ in $0<x<l$. Hence show that $\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}$.

UNIT V

19. Expand $f(x)=x-x^2$ as a Fourier series in $-l<x<l$ and using this series find the root mean square value of $f(x)$ in the interval.

Or

20. Compute the first three harmonics of the Fourier series of $f(x)$ given by the following table :

x :	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
$f(x)$:	1.0	1.4	1.9	1.7	1.5	1.2	1.0

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2014.

Third Semester

Common to all branches

MATHEMATICS — III

Time : Three hours Maximum : 75 marks

Answer ALL questions.

PART A — (10 × 2 = 20 marks)

1. When do you say a function $f(z)$ is analytic at a point $z=z_0$?

2. State the necessary conditions for $f(z)$ to be analytic in polar co-ordinates.

3. Obtain the Critical points of the transformation $w=z+\frac{a^2}{z}$.

4. Check whether $w=\frac{1}{z}$ is a bilinear transformation or not? Justify your result.

5. State Cauchy's integral theorem.

6. Find the formula for residue of $f(z)$ at $z=z_0$, a double pole.

7. Define a periodic function with period p.

8. State Dirichlet's conditions for the existence of Fourier series of $f(x)$.

9. Write down the formula for complex form Fourier series of $f(x)$ in $(0,2\pi)$.

10. State Parseval's theorem on Fourier series of $f(x)$ in $(-l, l)$.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE question from each Unit.

UNIT I

11. Determine the analytic function whose imaginary part is $\left(\frac{x-y}{x^2+y^2}\right)$.

Or

12. Show that the function $u(x, y)= e^{-x}(x\cos y + y\sin y)$ is harmonic and find its conjugate $v(x, y)$ such that $f(z)=u+iv$ is analytic.

UNIT II

13. Discuss the transformation $w=z^2$.

Or

14. Expand $f(z)=f(z)=\frac{z^2-1}{(z+2)(z+3)}$ in a Laurent's series if $2<|z|<3$.

UNIT III

15. Evaluate, using Cauchy's integral formula, $\int_C \frac{z+1}{(z^2+2z+4)}dz$ where C is the circle $|z+1+i|=2$.

Or

16. Evaluate $\int_0^{2\pi} \frac{d\theta}{1-2a\sin\theta+a^2}$ by using contour integration where $0<a<1$.

UNIT IV

17. Express $f(x)=(\pi-x^2)$ as a Fourier series of periodicity 2π in $0<x<2\pi$ and hence deduce $\sum_{n=1}^{\infty}\frac{1}{n^2}$.

Or

18. Find the Half Range cosine series of that function $f(x) = x(\pi - x)$ in the interval $0 < x < \pi$. Hence deduce that $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots = \frac{\pi^2}{90}$.

UNIT V

19. Compute upto first two Harmonics of the Fourier series of $f(x)$ given by the following table :

x :	0	T/6	T/3	T/2	2T/3	5T/6	T
$f(x)$:	1.98	1.3	1.05	1.3	−0.88	−0.25	1.98

Or

20. Find the complex form of Fourier series of cos ax in $(-\pi, \pi)$. Where a is not an integer.

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B.Tech. DEGREE EXAMINATION, APRIL/MAY 2015

Third Semester

Common to All Branches

MATHEMATICS – III

(For Students admitted from 2013–2014 Onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Verify whether the function $f(z) = \bar{z}$ is analytic function or not.

2. Show that $u = 2x - x^3 + 3xy^2$ is harmonic.

3. Find the map of the circle $|z| = 3$ under the transformation $w = 2z$.

4. Prove that the bilinear transformation has at most two fixed points.

5. Evaluate $\int_C \frac{z}{(z-1)(z-2)} dz$ where C is the circle $|z| = \frac{1}{2}$.

6. Calculate the residue of $f(z) = \frac{e^{2z}}{(z+1)^2}$ at its pole.

7. Write the condition for the function $f(x)$ to satify for the existence of a Fourier Series.

8. Obtain term constant a_0 of the fourier series for the function $f(x) = x^2, -\pi < x < \pi$.

9. Define the root mean square value of $f(x)$ over the interval (a, b).

10. What is meant by Harmonic Analysis?

SECTION B — (5 × 11 = 55 marks)

Answer All question, choosing ONE from each unit.

UNIT I

11. Determine the analytic function whose real part is $\frac{\sin 2x}{\cosh 2y - \cos 2x}$.

Or

12. Prove that $u = x^2 - y^2$ and $v = -y/x^2 + y^2$ are harmonic but $u + iv$ is not regular.

UNIT II

13. Find the bilinear transformation which maps the points $z = 0, -i, -1$ into $w = i, 1, 0$ respectively.

Or

14. Find the Image of $|z| = 2$ under the mapping (a) $w = z + 3 + 2i$ (b) $w = 3z$.

UNIT III

15. Using Cauchy's integral formula, evaluate $\int_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$ where C is $|z| = 3$.

Or

16. Evaluate $\int_0^{2\pi} \frac{\cos 2\theta \, d\theta}{5 + 4\cos\theta}$ using contour integration.

UNIT IV

17. Obtain the Fourier series of the periodic function defined by $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$ Deduce that $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots \infty = \frac{\pi^2}{8}$.

Or

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UNIT IV

17. Obtain the Fourier series for the function $f(x)$ given by $f(x)=\begin{cases}-x+1 & -\pi<x<0 \\ x+1 & 0<x<\pi\end{cases}$. Hence prove that $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots=\frac{\pi^2}{8}$. (11)

Or

18. Find the half range sine series of $f(x)=l-x$ in $(0,l)$.

Hence show that $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+\ldots\infty=\frac{\pi^2}{6}$. (11)

UNIT V

19. State and prove the Parseval's identity on Fourier series. (11)

Or

20. Find the Fourier series upto third harmonic. (11)

x:	0	$\pi/3$	$2\pi/3$	π	$4\pi/3$	$5\pi/3$	2π
y:	1	1.4	1.9	1.7	1.5	1.2	1

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2015.

Third Semester

MATHEMATICS — III

Common to All branches

(From 2013–14 onwards)

Time : Three hours   Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Test the analyticity for $f(z)=z^n$.

2. Prove that $v=3x^2y+x^2-y^3-y^2$ is harmonic.

3. Find the invariant points of $w=\frac{1+z}{1-z}$.

4. Define essential singularity with an example.

5. Evaluate $\int_C \frac{dz}{z-2}$ where C is $|z|=3$.

6. Find the residue of $f(z)=\frac{z^2}{(z-1)(z-2)^2}$ at its poles.

7. When does a function process a Fourier series expansion in terms of trigonometric terms?

If $f(x)=x^2+x$ is expressed as a Fourier series in the interval $(-2, 2)$ to which value this series converges at $x=2$?

If the Fourier series corresponding to $f(x)=x$ in the interval $(0, 2\pi)$ is $\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n \cos nx+b_n \sin nx)$ without finding the values of a_0, a_n and b_n find the value of $\frac{a_0^2}{2}+\sum_{n=1}^{\infty}(a_n^2+b_n^2)$.

Write down the complex form of the Fourier series of $f(x)$ in $(0, 2l)$ and the Euler's formula for the associated Fourier coefficient.

SECTION B — (5 × 11 = 55 marks)

Answer FIVE questions, choosing ONE from each Unit.

UNIT I

If $f(z)$ is a regular function of z, prove that $\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2=4|f'(z)|^2$. (11)

Or

Find the analytic function $f(z)=u+iv$ if $u+v=\frac{\sin 2x}{\cos 2x+\cosh 2y}$. (11)

UNIT II

13. (a) Prove that the region outside the circle $|z|=1$ maps onto the whole of the w-plane under the transformation $w=z+\frac{1}{z}$. (6)

(b) Find the bilinear transformation which maps the points $z=1, 2, 3$ onto $w=-1, \infty, 5$. (5)

Or

14. (a) Discuss the transformation $w=\sin z$. (5)

(b) Find the Laurent's series of $f(z)=\frac{1}{z(1-z)}$ in the region $1<|z+1|<2$. (6)

UNIT III

15. (a) Evaluate $\int_C \frac{e^{2z}}{(z-1)^4}dz$, Where C is the circle $|z-1|=1$, using Cauchy's integral formula. (6)

(b) Evaluate $\int_C \frac{4-3z}{z(z-1)(z-2)}dz$ where C is $|z|=\frac{3}{2}$, using residue theorem. (5)

Or

16. Using contour integration, evaluate $\int_{-\infty}^{\infty}\frac{x^2}{(x^2+4)(x^2+1)}dx$. (11)

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UNIT V

19. Find the Fourier series expansion of $f(x) = x^2$ in $-\pi < x < \pi$. Hence show that $\frac{1}{1^4} + \frac{1}{2^4} + \frac{1}{3^4} + \ldots\ldots\infty = \frac{\pi^4}{90}$.

Or

20. Compute the first two harmonic of the Fourier series of $f(x)$ from the following table

X	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
Y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

B.Tech. DEGREE EXAMINATION,
MAY 2016.

Third Semester

Civil Engineering

MATHEMATICS — III

(From 2013-14 batch onwards)

Time : Three hours Maximum : 75 marks

SECTION A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Show that the function $f(z) = \bar{z}$ is no where differentiable.

2. Show that $u = 2x - x^3 + 3xy^2$ is harmonic.

3. Find the bilinear map which maps the points $\infty, i, 0$ of plane onto $0, i, \infty$ of the plane.

4. Find the fixed points of the mapping $w = \frac{3-z}{1+z}$.

5. State Cauchy's integral theorem.

6. Find the residue of $f(z) = \frac{z}{(z-1)^2}$ at its poles.

7. State Dirichlet's conditions.

8. Find b_n the expansion of $f(x)=x^2$ as a Fourier series in $(-\pi, \pi)$.

9. Find the root mean square value of the function $f(x)=x$ in the interval $(0, l)$.

10. What do you mean by Harmonic analysis?

SECTION B — (5 × 11 = 55 marks)

Answer ALL questions choosing, ONE full question from each Unit.

UNIT I

11. If $f(z)$ is analytic function of z prove that $\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$.

Or

12. Prove that the function $v=e^{-x}(x\cos y + y\sin y)$ is harmonic and determine the corresponding analytic function $f(z)=u+iv$.

UNIT II

13. Expand $f(z)=\frac{z^2-1}{(z+2)(z+3)}$ as a Laurent's series if (a) $2<|z|<3$ (b) $|z|>3$.

Or

14. Find the bilinear transformation mapping the points $z=1, i, -1$ into the points $w=2, i, -2$.

UNIT III

15. Evaluate $\int_c \frac{\sin \pi z^2+\cos \pi z^2}{(z-1)(z-2)}dz$ where c is $|z|=3$.

Or

16. Evaluate $\int_0^\infty \frac{x^2}{(x^2+a^2)(x^2+b^2)}dx$, $a>0, b>0$.

UNIT IV

17. Expand $f(x)=x(2\pi-x)$ as a Fourier series in $(0, 2\pi)$. Hence show that $\frac{1}{1^2}+\frac{1}{2^2}+\frac{1}{3^2}+.....\infty=\frac{\pi^2}{6}$.

Or

18. Find the Fourier series $f(x)$ of with period $2l$ defined by $f(x)=\begin{cases} l-x, & 0<x<l \\ 0, & 0<x<2l \end{cases}$. Hence deduce that $1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+.....\infty=\frac{\pi}{4}$.